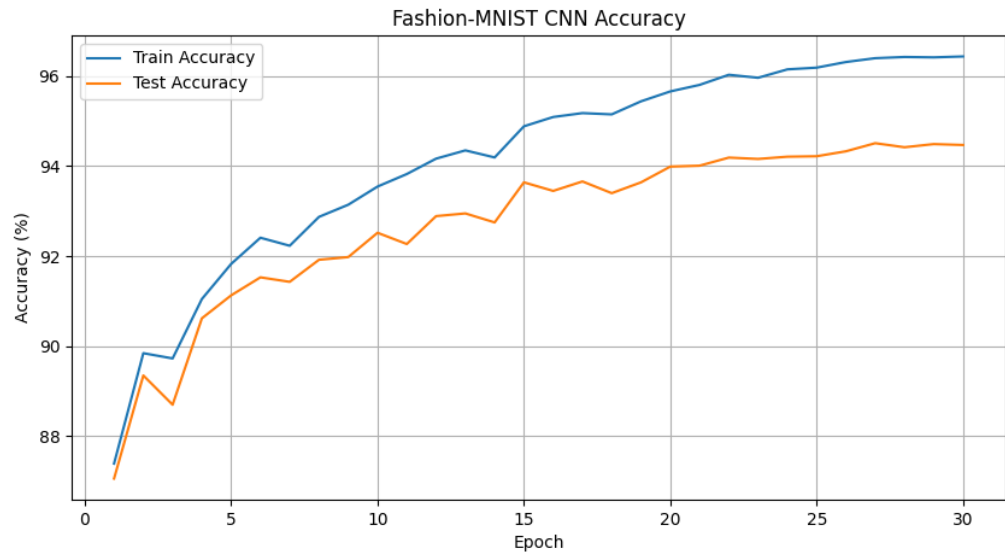


미니프로젝트 #1

202304178 구서연

202304330 한유정

1. PyTorch

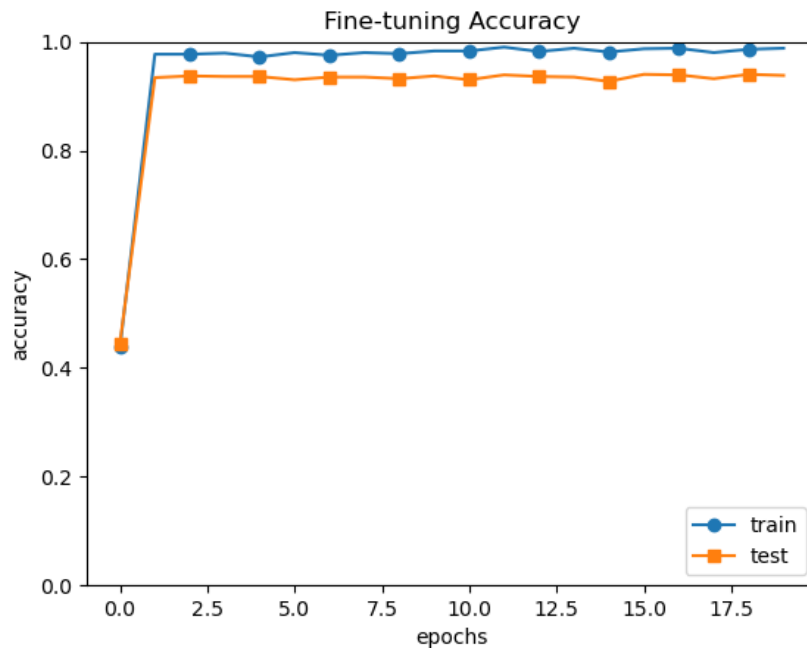


Epoch [11/30]	Train Acc(eval): 93.83%	Test Acc: 92.27%	Train Loss(eval): 0.1654	Test Loss: 0.2057	Train Acc(aug): 91.92%
Epoch [12/30]	Train Acc(eval): 94.17%	Test Acc: 92.89%	Train Loss(eval): 0.1567	Test Loss: 0.2003	Train Acc(aug): 92.26%
Epoch [13/30]	Train Acc(eval): 94.35%	Test Acc: 92.95%	Train Loss(eval): 0.1523	Test Loss: 0.1926	Train Acc(aug): 92.49%
Epoch [14/30]	Train Acc(eval): 94.19%	Test Acc: 92.75%	Train Loss(eval): 0.1532	Test Loss: 0.2010	Train Acc(aug): 92.59%
Epoch [15/30]	Train Acc(eval): 94.89%	Test Acc: 93.64%	Train Loss(eval): 0.1376	Test Loss: 0.1844	Train Acc(aug): 92.83%
Epoch [16/30]	Train Acc(eval): 95.09%	Test Acc: 93.45%	Train Loss(eval): 0.1350	Test Loss: 0.1872	Train Acc(aug): 92.97%
Epoch [17/30]	Train Acc(eval): 95.18%	Test Acc: 93.66%	Train Loss(eval): 0.1297	Test Loss: 0.1800	Train Acc(aug): 93.23%
Epoch [18/30]	Train Acc(eval): 95.15%	Test Acc: 93.40%	Train Loss(eval): 0.1307	Test Loss: 0.1841	Train Acc(aug): 93.48%
Epoch [19/30]	Train Acc(eval): 95.44%	Test Acc: 93.64%	Train Loss(eval): 0.1220	Test Loss: 0.1792	Train Acc(aug): 93.63%
Epoch [20/30]	Train Acc(eval): 95.66%	Test Acc: 93.99%	Train Loss(eval): 0.1171	Test Loss: 0.1711	Train Acc(aug): 93.94%
Epoch [21/30]	Train Acc(eval): 95.81%	Test Acc: 94.01%	Train Loss(eval): 0.1137	Test Loss: 0.1739	Train Acc(aug): 93.98%
Epoch [22/30]	Train Acc(eval): 96.03%	Test Acc: 94.19%	Train Loss(eval): 0.1085	Test Loss: 0.1692	Train Acc(aug): 94.20%
Epoch [23/30]	Train Acc(eval): 95.96%	Test Acc: 94.16%	Train Loss(eval): 0.1078	Test Loss: 0.1697	Train Acc(aug): 94.27%
Epoch [24/30]	Train Acc(eval): 96.15%	Test Acc: 94.21%	Train Loss(eval): 0.1061	Test Loss: 0.1672	Train Acc(aug): 94.42%
Epoch [25/30]	Train Acc(eval): 96.19%	Test Acc: 94.22%	Train Loss(eval): 0.1037	Test Loss: 0.1671	Train Acc(aug): 94.60%
Epoch [26/30]	Train Acc(eval): 96.31%	Test Acc: 94.33%	Train Loss(eval): 0.1021	Test Loss: 0.1663	Train Acc(aug): 94.62%
Epoch [27/30]	Train Acc(eval): 96.40%	Test Acc: 94.51%	Train Loss(eval): 0.0997	Test Loss: 0.1632	Train Acc(aug): 94.65%
Epoch [28/30]	Train Acc(eval): 96.42%	Test Acc: 94.42%	Train Loss(eval): 0.0994	Test Loss: 0.1635	Train Acc(aug): 94.80%
Epoch [29/30]	Train Acc(eval): 96.42%	Test Acc: 94.49%	Train Loss(eval): 0.0990	Test Loss: 0.1633	Train Acc(aug): 94.82%
Epoch [30/30]	Train Acc(eval): 96.44%	Test Acc: 94.47%	Train Loss(eval): 0.0989	Test Loss: 0.1634	Train Acc(aug): 94.78%

Best Epoch: 27
Best Test Accuracy: 94.51%

Final Evaluation with Best Model
Final Train Loss: 0.0997
Final Train Accuracy: 96.40%
Final Test Loss: 0.1632
Final Test Accuracy: 94.51%

2. Scratch



```
train loss:0.03921802686992301
train loss:0.05784587574072095
train loss:0.06842961514324222
train loss:0.05177974775697686
train loss:0.08386671187503944
train loss:0.09838149987995522
train loss:0.06580937848381298
train loss:0.051068506032025535
train loss:0.10466488821681193
train loss:0.0650200532361694
train loss:0.08925574299275311
train loss:0.11407114701546561
train loss:0.09497713611617223
train loss:0.08718448960614854
train loss:0.06260476215388364
train loss:0.05634335820500915
train loss:0.06700290234596362
train loss:0.06603535126322688
train loss:0.0898801706630188
train loss:0.07115288374250833
train loss:0.06099288247040042
train loss:0.04226773864792291
train loss:0.07935575385027219
train loss:0.08444515116688005
train loss:0.106440285013382
train loss:0.042832245575745505
train loss:0.07050122422233751
train loss:0.038619876139921086
train loss:0.07526277892526968
train loss:0.09945109564899221
train loss:0.07081893244063957
train loss:0.10148826281028595
```

```
===== Final Test Accuracy =====
test acc:0.9287
Saved Network Parameters!
```

```
(base) C:\Users\User\Desktop\학교\universitiy\26-1\인공지능개론>
```

```
train loss:0.07990622815895722
train loss:0.08336920050520344
train loss:0.07721031167222153
train loss:0.06097620038551326
train loss:0.04980908096023434
train loss:0.08572669069148728
train loss:0.08623728859987279
train loss:0.0895106635328938
train loss:0.07539490075686858
train loss:0.07591761768723887
train loss:0.10853410161667362
train loss:0.04153948627934673
train loss:0.06314030159441655
train loss:0.0455655666341174
train loss:0.05736367557163368
train loss:0.06361530942878896
train loss:0.07210779023799829
train loss:0.06040573484002151
train loss:0.09713755066468945
train loss:0.1185579891159518
train loss:0.06276886169671841
train loss:0.09674615669893696
train loss:0.07525554081797181
train loss:0.05780873454919927
train loss:0.060925492617842227
train loss:0.10459121233898366
train loss:0.08561114428508165
train loss:0.08334207329190885
train loss:0.060954559505892236
train loss:0.09619985627601853
train loss:0.07065632265115852
train loss:0.09124553969665505
train loss:0.10773213445391433
train loss:0.06828777832287675
train loss:0.05329436445965538
```

```
=== epoch:20, train acc:0.988, test acc:0.938 ===
```